Mark schemes

Q1.

(a) (i) $\mathbf{a} = \frac{1}{4}(8\mathbf{i} - 3\mathbf{j}) = 2\mathbf{i} - 3\mathbf{j}$

Use of $\boldsymbol{F} = m\boldsymbol{a}$*, must be applied to both components*

M1

Correct acceleration

A1

2

(ii) $\mathbf{v} = 20(2\mathbf{i} - 3\mathbf{j}) = 40\mathbf{i} - 60\mathbf{j}$

Use of $\boldsymbol{v} = \boldsymbol{u} + \boldsymbol{a}t$ *with* $\boldsymbol{u} = 0\boldsymbol{i} + 0\boldsymbol{j}$

M1

Correct $\boldsymbol{v}$

A1

2

(iii) $\mathbf{r} = \frac{1}{2}(2\mathbf{i} - 3\mathbf{j}) \times 20^2$

Use of constant acceleration equation to find $\boldsymbol{r}$ *with* $\boldsymbol{u} = 0\boldsymbol{i} + 0\boldsymbol{j}$

M1

Correct expression

A1

$= 400\mathbf{i} - 600\mathbf{j}$

Correct $\boldsymbol{r}$ *in simplified form*

A1

3

(b) $\mathbf{r} = 400\mathbf{i} - 600\mathbf{j} + 25(40\mathbf{i} - 60\mathbf{j})$

Use of $\boldsymbol{r} + 25\boldsymbol{v}$

M1

Correct expression

A1ft

$= 1400\mathbf{i} - 2100\mathbf{j}$

Correct position vector

A1ft

$r = \sqrt{1400^2 + 2100^2}$ = 2520 m (to 3 sf)

Finding magnitude

m1

Correct distance from correct working

A1ft

5

Alternative: *Straight line method*
M1 two distances r and 25v
A1 for each distance
m1 adding
A1 correct final answer

[12]

Q2.

(a) $v = 0 + 0.5 \times 10 = 5$ ms^{-1}

Use of v = u + at

M1

Correct v

A1

2

(b) $s = \frac{1}{2} \times 0.5 \times 10^2 = 25$ m

Use of a constant acceleration equation to find s

M1

Correct s

A1

2

(c) $F - 40000 = 200000 \times 0.5$

Three term equation of motion

M1

Correct equation

A1

$F = 140000$ N

Correct force

A1

3

[7]

Q3.

(a) $R = 60 \times 9.8 \times \cos 40° = 450$ N

Resolving perpendicular to the plane

M1

Correct reaction force

A1

2

(b) $F = 60 \times 9.8 \times \cos 50° = 378$

Resolving parallel to the plane

M1

Correct F

A1

$F \leq \mu R$

Use of $F \leq \mu R$ or $F = \mu R$

m1

$\mu \geq 0.839$

Correct inequality
Accept 0.840

A1

4

(c) $F = 0.2 \times 60 \times 9.8 \cos 40°$

$60a = 60 \times 9.8 \cos 50° - 0.2 \times 60 \times 9.8 \cos 40°$

Use of $F = \mu R$ to find F but not with R = 60g

M1

Three term equation of motion

M1

Correct equation

A1

$a = 4.80$ ms^{-2}

Correct acceleration

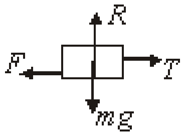
A1

5

[11]

Q4.

(a)



Correct diagram

B1

1

(b) $R = 4 \times 9.8 = 39.2$ N

Correct normal reaction

B1

1

(c) $T - 4 \times 39.2 = 4 \times 2$

Three term equation of motion

M1

Correct equation

A1

$T = 23.7$ N (to 3sf)

Correct T

A1

3

(d) $20 - 0.4 \times 39.2 = 4a$

M1

$a = 1.08$ ms^{-2}

A1

2

[7]

Q5.

(a) $12g - T = 12a$

Equation of motion for one particle

M1

Correct equation

A1

$T - 8g = 8a$

Equation of motion for other particle

M1

Correct equation

A1

$8a + 8g = 12g - 12a$

$20a = 4g$

$a = \frac{4g}{20} = 1.96\ \text{ms}^{-2}$

ag *Correct a from correct working*

A1

5

(b) $T = 8 \times 1.96 + 8 \times 9.8$

$= 94.1$ N

Substituting value for a into equation of motion to find T

M1

Correct T

A1

2

(c) $7 = 0 + 1.96t$

Using constant acceleration to find t

M1

$t = \frac{7}{1.96} = 3.57\ s$ (to 3 sf)

Correct t

A1

2

[9]

Q6.

(a) $s = \frac{1}{2}(40 + 60) \times 4$

Equation based on area of trapezium of rectangle plus two triangles

M1

Correct equation

A1

= 200m

Correct final distance

A1

3

(b) $a = \frac{4}{5} = 0.8 \text{ ms}^{-2}$

Attempting to find acceleration

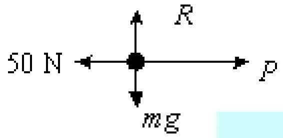
M1

Correct acceleration

A1

2

(c) (i)



Correct force diagram

B1

1

(ii) $P - 50 = 65 \times 0.8$

Three term equation of motion

M1

Correct equation

A1

$P = 102$ N

Correct value

A1

3

[9]

Q7.

(a) $0.8 = \frac{1}{2} a \times 2^2$

Constant acceleration equation with u = 0

M1

Correct equation

A1

a = 0.4 ms^{-1}

Correct acceleration

A1

3

(b) $2 \times 0.4 = 2 \times 9.8 - T$

Three term equation of motion

M1

Correct equation

A1

$T = 2 \times 9.8 - 2 \times 0.4 = 18.8$ N

Correct tension

A1

3

(c) $0.4m = 18.8 - 9.8\,m$

Three term equation of motion

M1

Correct equation
Solving for m

A1

$m = \dfrac{18.8}{10.2} = 1.8\text{kg}$

M1

Correct m (2sf)

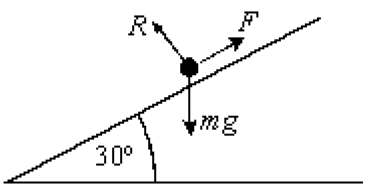
A1

4

[10]

Q8.

(a)



Correct force diagram

B1

1

(b) $R = 5 \times 9.8 \cos 30°$

Finding normal reaction

M1

$F = 0.2 + 5 \times 9.8 \cos 30° = 8.5$ N (to sf)

Finding friction

M1

Correct friction

A1

3

(c) $5 \times 9.8 \sin 30° - 0.2 \times 5 \times 9.8 \cos 30° = 5a$

Three term equation of motion

M1

Correct equation

A1

$$a = \frac{5 \times 9.8 \sin 30° - 0.2 \times 5 \times 9.8 \cos 30°}{5}$$

Solving for a

M1

$= 3.20 \text{ ms}^{-2}$

Correct a

A1

4

(d) $v^2 = 0^2 + 2 \times 3.203 \times 1.2$

Constant acceleration equation with u = 0

M1

Correct equation

A1 ft.

$v = 2.77 \text{ ms}^{-1}$

A1 ft.

3

[11]

Q9.

(a) $1000\dfrac{dv}{dt} = 2000 - 40v$

Use of Newton's 2nd law

M1

$\dfrac{dv}{dt} = \dfrac{2000 - 40v}{1000} = \dfrac{50 - v}{25}$

Correct expression from correct working

A1

2

(b) $\displaystyle\int \frac{1}{50 - v}\,dv = \int \frac{1}{25}\,dt$

$-\ln|50 - v| = \dfrac{t}{25} + c$

Integration to obtain t and 1n term

M1

Correct integration

A1

$50 - v = Ae^{\frac{t}{25}}$

$v = 50 - Ae^{-\frac{t}{25}}$

Solving for v

m1

$v = 0, t = 0 \Rightarrow A = 50$

Finding constant of integration.

m1

$v = 50\left(1 - e^{-\frac{t}{25}}\right)$

Correct final answer

A1

5

[7]

Q10.

(a) 120 = $\frac{1}{2}a \times 50^2$

Use of constant acceleration equation

M1

Correct equation

A1

$a = \frac{240}{2500} = 0.096\ \text{ms}^{-2}$

Correct acceleration from correct working

A1

3

(b) $v^2 = 2 \times 0.096 \times 200$

Use of constant acceleration equation

M1

Correct equation

A1

$v = 6.20\ \text{ms}^{-1}$

Correct velocity

A1

3

(c) 200 = $\frac{1}{2} \times 0.096t^2$

Use of a constant acceleration equation

M1

Correct equation

A1

$t = \sqrt{\frac{400}{0.096}} = 64.5\ \text{s}$

Correct time

A1

3

(d) F – 250 × 9.8 = 250 × 0.096

Three term equation of motion

M1

Correct equation

A1

F = 2470 N (to 3sf)

Correct force

A1

3

[12]

Q11.

(a) $15^2 = 20^2 + 500a$

Use of constant acceleration equation

M1

Correct equation

A1

$a = -0.35$ ms^{-2}

Correct acceleration from correct working

A1

3

(b) $P - 1200 \times 9.8 \cos 86° = 1200 \times (-0.35)$

Three term equation of motion

M1

Correct equation

A1

Solving for P

m1

$P = 400$ N

Correct P

A1

4

(c) $P = 400 + 300 = 700$ N

Including resistance force

M1

Correct value

A1

2

(d) Expect air resistance force to decrease as car moves up the slope as it is slowing down

Air resistance
Describing variation

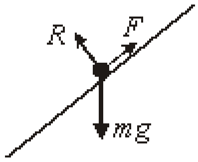
B1

2

[11]

Q12.

(a)



Correct force diagram
No extra forces

B1

1

(b) (i) $R = 80 \times 9.8 \cos 30° = 679.0$ N

Resolving perpendicular to the slope
Correct Reaction force

M1 A1

2

(ii) $F = 0.3R = 204$ N

Multiply by 0.3
***ag** Correct answer obtained*

M1 A1

2

(c) $80a = 80 \times 9.8 \sin 30° - 203.7$

Three term equation of motion
Correct equation

M1 A1

$a = 2.35$ ms^{-2}

Solving for a
awrt 2.35

m1 A1

4

[9]

Q13.

(a) (i) $60 \times 9.8 - T = 60a$

Equation of motion for one boy

Correct equation

M1 A1

$T - 40 \times 9.8 = 40a$

Equation of motion for other boy

Correct equation

M1

$588 - (40a + 392) = 60a$

$a = 1.96\ \text{ms}^{-2}$

ag Correct answer from correct working

A1

5

(ii) $T = 40 \times 1.96 + 392 = 470$ N (to 3 sf)

Substituting for a and solving for T

awrt 470

M1 A1 M1

2

(b) $v^2 = 0^2 + 2 \times 1.96 \times 2 = 7.84$

Finding v^2 *using s = 2, u = 0 and a = 1.96*

Correct v or v^2

M1 A1

$0^2 = 7.84 + 2 \times (-9.8)s$

Equation for s *with* v *= 0 and a = ±g*

m1 A1

$s = 0.4$

Solving for s

m1

$h_{max} = 2 + 0.4 = 2.4$ m

Solving for s

Obtaining 0.4

Adding 2 to get 2.4

m1 A1 A1

7

[14]

Q14.

(a) $m\frac{dv}{dt} = -mkv^3$

Forming a differential equation

M1

$\int \frac{1}{v^3}\,dv = -\int k\,dt$

$-\frac{1}{2v^2} = -kt + c$

Integrating to get a $\frac{1}{v^2}$ *term*

Correct Integral including c

m1 A1

$v = U,\ t = 0 \Rightarrow c = -\frac{1}{2U^2}$

Finding c

Correct c

m1 A1

$\frac{1}{2v^2} = kt + \frac{1}{2U^2} = \frac{2ktU^2 + 1}{2U^2}$

$v^2 = \frac{U^2}{2ktU^2 + 1}$

Correct final answer from correct working

A1

6

(b) v tends to zero

Allow decreases

B1

1

[7]

Q15.

(a) $34 = 8 + 20a$

Using constant acceleration equation
Correct equation

M1 A1

a = 1.3 ms^{-2}
Correct a

A1

3

(b) $s = \frac{1}{2} \times (8 + 34) \times 20 = 420$m
Using constant acceleration equation
Correct distance

A1

2

(c) $F - 400 = 1250 \times 1.3$
Three term equation of motion
Correct equation

M1 A1

F = 2025 N
Correct force

A1

3

[8]

Q16.

(a) $\mathbf{F} = (2 + 6 - 2)\mathbf{i} + (3 - 8 + 10)\mathbf{j}$
Summing the forces

M1

$= 6\mathbf{i} + 5\mathbf{j}$
Correct resultant

A1

2

(b) $\mathbf{a} = \frac{1}{8}(6\mathbf{i} + 5\mathbf{j}) = 0.75\mathbf{i} + 0.625\mathbf{j}$
Dividing by 8 to get correct value

B1

1

(c) (i) $\mathbf{v} = (3\mathbf{i} - 2\mathbf{j}) + 40(0.75\mathbf{i} + 0.625\mathbf{j})$

use of $\mathbf{v} = \mathbf{u} + \mathbf{a}t$

M1

$= 33\mathbf{i} + 23\mathbf{j}$

Correct $\mathbf{v}$

A1

2

(ii) $\mathbf{r} = (3\mathbf{i} - 2\mathbf{j}) \times 40 + \frac{1}{2}(0.75\mathbf{i} + 0.625\mathbf{j}) \times 40^2$

Vector constant acceleration equation to find $\mathbf{r}$

Correct equation

M1 A1

$= 720\mathbf{i} + 420\mathbf{j}$

Correct position vector

A1

$r = \sqrt{720^2 + 420^2} = 834$ m

Finding magnitude

Correct magnitude

M1 A1

5

[10]

Q17.

(a) $5^2 = 1^2 + 2 \times 4a$

Using constant acceleration equation to find a

M1 A1

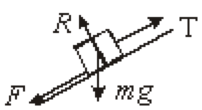
$a = 3 \text{ ms}^{-2}$

Correct equation

A1

3

(b) (i)



Correct force diagram

B1

1

(ii) $R = 20 \times 9.8 \cos 30° = 170$ N

Finding the normal reaction

M1

$F = 0.4 \times 20 \times 9.8 \cos 30° = 67.9$ N

Using F = μ R

M1 A1

3

(iii) $T - 67.9 - 20 \times 9.8 \sin 30° = 20 \times 3$

Four term equation of motion

Correct equation

M1 A1

$T = 226$ N

Solving for T

Correct T

m1 A1

4

[11]

Q18.

(a) $20^2 = 12^2 + 2 \times a \times 200$

Using a constant acceleration equation with a unknown

M1

$a = \dfrac{400 - 144}{400} = 0.64 \text{ ms}^{-2}$

Correct equation

Correct result from correct working

A1/A1

3

(b) $20 = 12 + 0.64t$

Using a constant acceleration equation with t *unknown*

M1

$t = \frac{20 - 12}{0.64} = 12.5$ seconds

Correct equation

Correct result from correct working

A1/A1

3

(c) (i) $F = 1200 \times 0.64 = 768$ N

Correct answer only

B1

1

(ii) $F - 450 = 768$

Three term equation of motion

M1

$F = 1220$ N

Correct result

Accept 1218

A1

2

[9]

Q19.

(a) $6060 - 600 \times 9.8 = 600a$

Three term equation of motion

M1

$a = \frac{6060 - 5880}{600} = 0.3 \text{ ms}^{-2}$

Correct acceleration from correct working

A1

2

(b) $8 = \frac{1}{2} \times 0.3t^2$

Using a constant acceleration equation with t *unknown*

M1

$t = \sqrt{\frac{16}{0.3}} = 7.30$ seconds

Correct equation
Correct result
Accept 7.3

A1/A1

3

(c) $v^2 = 2 \times 0.3 \times 8$

Using a constant acceleration equation with v unknown and u = 0

M1

$v = \sqrt{4.8} = 2.19 \text{ ms}^{-1}$

Correct velocity

A1

2

[7]

Q20.

(a) $T - 0.8 \times 5 \times 9.8 = 5a$

Equation of motion for one particle

M1

$T = 5a + 39.2$

Correct equation

A1

$3 \times 9.8 - T = 3a$

Equation of motion for other particle

M1

$29.4 - (5a + 39.2) = 3a$

Correct equation

A1

$-9.8 = 8a$

Solving for a

M1

$a = -1.225 \text{ ms}^{-2}$

Correct a

A1

6

(b) (ii) $T = 5a + 39.2$

Substituting and solving for T

M1

$T = 33.1$N

Correct T

A1

2

(c) $0^2 = 2^2 + 2 \times (-1.225)s$

Use of CA equation with v *= 0*

Correct equation (allow a *= 1.225)*

M1/A1

$s = \frac{4}{2.45} = 1.63$ m

Correct answer from correct working

(allow 1.64)

A1

3

[11]